Investigating Metrics for a Development Effort Prediction Model of Web Applications

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Abstract

Although in the hypermedia literature there are metrics proposed to measure hypermedia processes and products, many lack the necessary theoretical and empirical validation.

To address these issues, this paper presents the results of a quantitative case study which validated empirically a set of metrics proposed to measure the development effort involved in authoring Web applications. These metrics adhere to the representational theory of measurement.

The results obtained for the case study have shown that three out of four of the proposed metrics presented statistically significant correlations with development effort, suggesting that they may be useful parameters for a prediction model that estimates the effort involved in developing Web applications.

1. Introduction

Developing good quality hypermedia applications has a high cost, mostly in time and the amount of difficulty involved for the authors [4]. Therefore, if we know more about how those applications are developed, we can have better control over them and, consequently, the chances of improving the development will increase. To facilitate control, measurement and feedback are obtained as the first step to improve products, processes and resources.

As engineering involves the analysis of measurements [18], hypermedia engineering and Web engineering, will only become true engineering disciplines when a solid foundation of measurement-based theories is built.

Although there is some research into hypermedia metrics, most of it to date uses an analytical method of research without further empirical justification. Consequently, many of the published research findings can only be interpreted as analytical advocacy research [8].

The main objective of this research project was to investigate the process of Web authoring by using a scientific method of research, which comprised the following steps:

i) The proposal of metrics to measure the maintainability of Web applications, the reusability of Web documents and the development effort involved in authoring a Web application.

ii) The proposal of metrics which adhere to the representational theory of measurement.

iii) The empirical and theoretical validation of our metrics according to the validation framework proposed by Kitchenham et al. [12].

The first part of our research project was a survey, which measured the maintainability of hypermedia applications and the reusability of hypermedia information [15]. This paper presents the second part of the project - a quantitative case study aimed at validating empirically our proposed metrics as possible measures of the development effort involved in authoring Web applications. In Section 2, the quantitative case study is described. Section 3 presents the validity of the case study evaluation and the theoretical validation of the proposed metrics described in Section 4. Finally, in Section 5, the results of the case study are presented and in Section 6 we give our conclusions and comments on future work.

2. The Case Study Evaluation

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2.1 Design of the Evaluation

Two questionnaires were used in the case study: the first one - Q1 - asked subjects to rate their level of Web authoring experience on a scale from 1 (little experience) to 10 (a great deal of experience); the second one - Q2 - (see Appendix A) was used to measure the independent and dependent variables. Both questionnaires were checked by members of the research group for ambiguous questions, unusual tasks, definitions in the appendix and the number of questions.

The subjects who participated in the evaluation were 45 second-year Computer Science students attending the Human-Computer Interaction course. An analysis of the questionnaires obtained showed that six were missing fundamental information. Consequently, these samples were not used in the data analysis.

Subjects were given a coursework that consisted of developing Web applications to teach about Human-Computer Interaction concepts. They were also required to structure their applications according to the Cognitive Flexibility Theory (CFT) >22] principles, and to create the applications using a minimum of 50 documents.

In order to reduce any learning effects (confounding factors) >11] subjects were given an initial task - to develop their personal Web pages - Two Web tutorials were suggested so that it would be possible for subjects with no Web authoring experience to learn how to develop Web pages. In addition, all the subjects received training on the CFT authoring principles (90 minutes).

The questionnaire Q2 was handed in together with the coursework (the Web application). This questionnaire had two sections: one to collect attributes of the application and another to collect the elapsed time spent in developing the application. The development was split in several sub-tasks along the development life-cycle, adapted from existing literature >1,9,10,13,14,17,20,21,23] and organised as follows: i) Structure the application; ii) Create the text contents; iii) Create the graphics contents; iv) Create the video contents; v) Create the audio contents; vi) Create the links; vii) Design the interface: planning of the interface; viii) Design the interface: implementing of the interface; ix) Test the application: test the links in the application; and x) Test the application: testing the media.

Each of these subtasks would correspond to an elapsed time, which would be correlated statistically with the independent variables in order to check the hypotheses proposed.

2.2 Dependent and Independent Variables

The independent variables (metrics) used in the case study evaluation were: hyperdocument size, connectivity, perceived compactness and perceived stratum; The dependent variable used to measure the development effort was the estimated elapsed time it took each subject to develop the Web application. At this stage of the research our concern was with the measurement for assessment rather than a measurement for prediction >7]. This means that we were interested in understanding the development effort involved in authoring Web applications and how the independent variables were related to it, rather than producing a mathematical model (prediction model) to be used as a predictor of development effort.

Hyperdocument size refers to the number of documents that each Web application has. Documents are HTML files, and a Web application is a collection of documents designed with a specific aim. In our case, the domain considered to be representative was education. The Web applications considered are 'static', where the number of dynamically generated pages is quite small, if not absent.

Connectivity refers to the number of links that the Web application has. The links considered here are either structural or referential links >5], leaving apart dynamically computed links.

Compactness >3] indicates how inter-connected the documents are. A completely connected application means that, from each document, there are links reaching all the other documents in the application. Compactness was measured in the case study evaluation by asking subjects to give an estimate of it, from 1 (completely disconnected) to 5 (completely connected). Consequently, the compactness measured was the perceived compactness (P-Compactness).

Stratum >3] indicates to what degree the Web application is organised for directed reading. A sequential application represents an application where there is only one path to be followed by the reader. For the case study evaluation, subjects were asked to estimate the stratum of the application, from 1 (no sequential navigation) to 5 (sequential navigation).

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Consequently, the stratum measured was the perceived stratum (P-Stratum).

2.3 The Hypotheses

Although in a scientific method of research the hypotheses to be tested should be the null hypotheses, we decided to present the alternative hypotheses as they explicitly show the direction of the relationship between the independent and dependent variables. Our stated hypotheses for the evaluation were as follows:

H1 - The bigger the hyperdocument size the longer it takes to develop the Web application.

H2 - The greater the connectivity the longer it takes to develop the Web application.

H3 - The higher the p-compactness the longer it takes to develop the Web application.

H4 - The smaller the p-stratum the longer it takes to develop the Web application.

3. Validity of the Case Study Evaluation

There are three types of threats to the validity of an empirical study >11]:

- Construct validity: Represents to what extent the independent and dependent variables precisely measure the concepts they claim to measure.
- Internal validity: Represents to what extent conclusions can be drawn about the casual effect of the independent variables on the dependent variables.
- External validity: Represents the domain to which a study's findings can be generalised.

3.1 Construct Validity

- subjects, rather than directly measured in the applications. It can be argued that estimated values are not ideal and may weaken the results obtained. However, it would not be feasible to ask subjects to calculate themselves those measures as it would be very time-consuming. Compactness, as proposed by Botafogo et al. >3], also uses an estimated value on its calculation (the conversion constant), leading to subjective measures as well. Finally, stratum, as proposed by Botafogo et al. >3] is described by the authors as "not a perfect metric, where some problems are evident". Therefore, we decided to re-use these concepts and measure them in a different way.
- Although the number of documents has been identified by hypermedia authors as a valuable metric to measure hyperdocument sLze >6], we are aware that there are other metrics which can also be used (for example, the length of the application, the complexity of documents and the functionality offered) and they will be the focus of our next evaluation.

3.2 Internal Validity

The following problems might threaten the internal validity of the experiment:

- There were three confounding factors in this evaluation: i) subjects' experience; ii) maturation effects, which are learning effects caused by subjects learning as an experiment proceeds; and iii) tools used to help develop/manage the Web application. The data collected showed that: i) the experiences did vary greatly. Consequently, it was decided to consider two separate groups - subjects with experience levels ranging from 1 to 5 (22 subjects) - LEL group - and subjects with experience levels ranging from 6 to 10 (17 subjects) - HEL group; ii) subjects were asked to develop their Web pages before developing the applications and they all received training in the CFT principles. In addition, they were given a general structure for their applications, using the CFT. Consequently, it seems that the maturation effects were controlled, or at least reduced; and iii) the number of tools used did not vary greatly. This result was confirmed statistically.
- The selection effects, which are due to natural variations in subject performance, were controlled.
- The instrumentation effects in general did not occur in this evaluation. The questionnaires used were the same and the same hypermedia system was used to develop the applications.

3.3 External Validity

The following problems might limit the external validity of the experiment:

- The results may be domain dependent as all the subjects answered the questionnaires based on

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their experience in developing Web applications for education. This evaluation should therefore be repeated in domains other than education if the results are to be generalised to other domains.

- The experimental scale >19] is a threat when the experimental setting or the materials are not representative of industrial practice. The applications developed presented good quality interface and contents, when compared to other Web applications found in industry. Therefore, from this perspective it seems that the applications developed are representative of industrial practice.
- The applications did not have more than 100 documents and 300 links, on average, as the subjects were supposed to use only three working days to develop them. These numbers represent small applications and are probably not representative of the large Web applications for education developed by some organisations, in general. On the other hand, the applications developed had a similar to or even better interface and contents quality than Web applications developed by professionals, as noticed by myself when marking the Web applications developed by the students. These results seem to indicate that the results of the case study are likely to scale-up.
- A threat to subject generalizability >19] may exist when the subject population is not drawn from the wider population. The subjects that participated in the case study may not be representative of Web professionals, since second-year Computer Science students cannot be categorised as experienced Web authors. The use of students as subjects, while sometimes considered unrealistic, is justified for two reasons: firstly, empirical evidence by Bohem-Davis and Ross (1992) indicates that students are equal to professionals in many quantifiable measures, including their approach to developing software; secondly, for pragmatic considerations, having students as subjects was the only viable option for this case study.

3. Theoretical Validation of the Proposed Metrics

This Section presents a theoretical validation of our metrics according to the validation framework proposed by Kitchenham et al. >12].

For the validation framework, to decide whether a measurement is valid it is necessary to confirm:

- Attribute validity, i.e., whether the attribute is actually exhibited by the entity being measured.
- Unit validity, i.e., whether the measurement unit being used is an appropriate means of measuring the attribute.
- Instrument validity, i.e., whether any model underlying a measuring instrument is valid and the measuring instrument is properly calibrated.
- Protocol validity, i.e., whether an acceptable measurement protocol is adopted.

The properties of measures used in the validation are:

(1) For an attribute to be measurable, it must allow different entities to be distinguished from one another.
(2) A valid measure must obey the representation condition, preserving our intuitive notions about the attribute and the way in which it distinguishes different entities.
(3) Each unit of an attribute contributing to a valid measure is equivalent.
(4) Different entities can have the same attribute value.

The theoretical validation for our metrics is presented in table 1:

4.1 Validation of Instrument Models

Validating a measurement instrument represents the confirmation that the instrument accurately measures attribute values in a given unit. The instrument used in the quantitative case study measured all the attributes using appropriate scales and units.

4.2 Validation of Measurement Protocols

The protocols applied to the instrument used was validated by peer acceptance. , In addition, when asking for volunteers, the questionnaire and evaluation's objectives were always explained.

3. The Results of the Case Study

All the correlations presented in this section were calculated by using the non-parametric Spearman's

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rank correlation test with a 1-tailed significance test at the 1% and 5% levels.

Statistically significant negative correlations for P-Stratum and statistically significant positive correlations for Hyperdocument Size, Connectivity and P-Compactness helped to confirm our hypotheses.

The legend for the subtasks presented in tables 4 and 5 is described below (see figure 1):

Acronym	Description	Acronym	Description
Createt	Create the text contents	Structap	Structure the application
Createg	Create the graphics contents	Designip	Planning of the interface
Createv	Create the video contents	Designii	Implementing of the interface
Createa	Create the audio contents	Testl	Testing the links in the application
Createc	Total time creating text, graphics, video, audio	Testm	Testing the media.
Createl	Create the links		

Figure 1. Legend for the subtasks involved in the development of a hypermedia application

Table 1. Theoretical validation of the direct measures

Metric	Scale	Unit	Properties that apply
Hyperdocument size	absolute	documents	(1) Different applications may have different hyperdocument sizes. (1) The same application with different names will lead to the same hyperdocument size if a software program is used as an instrument. (2) Has one unit. (3) Different applications can have the same hyperdocument size.
Connectivity	absolute	links	(2) Different applications may have different connectivity. (3) The same application with different names will lead to the same connectivity if a software program is used as an instrument. (4) Has one unit. (5) Different applications can have the same connectivity.
Compactness	ordinal	Ranking of cross-referencing	(1) Different applications may have different compactness. (2) The same application with different names may lead to the same compactness if the respondents share a common knowledge about the application's structure. (3) Has one unit (4) Different applications can have the same connectivity
Stratum	ordinal	Ranking of linearity	(1) Different applications may have different stratum. (2) The same application with different names may lead to the same stratum if the respondents share a common knowledge about the application's structure. (3) Has one unit. (4) Different applications can have the same stratum.
Link generality	Nominal	Boolean (anchor/not anchor)	(1) Different systems may have different link generalities. (2) The same system with different names will lead to the same link generality. (3) Has one unit. (4) Different systems can have the same link generality.
Development effort for a hypermedia application	Absolute	Estimated elapsed time in hours	(1) Different applications may have different development efforts (duration). (2) The same hypermedia application with different names may lead to the same duration, if answered by the same author. (3) Has one unit. (4) Different hypermedia applications can require the same development effort.

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For the LEL subjects, considering the results presented in table 2, it seems that, as far as the development effort involved in authoring a Web applications is concerned:

- The bigger the htyperdocument size the more time-consuming the development of the Web application becomes (confirms hypothesis H1).
- The greater the connectivity the more time-consuming the development of the Web application becomes (confirms hypothesis H2).
- The higher the p-compactness the more time-consuming the development of the Web application becomes (confirms hypothesis H3).

The elapsed time spent in creating the text documents (see table 3) showed a statistically significant negative correlation with p-stratum, suggesting that it took longer to create the text documents of a Web application developed with a less linear style (concerning its navigation). However, this result was not sufficient to confirm hypothesis H.4.

For the HEL subjects, considering the results obtained, it seems that, as far as the development effort involved in authoring a Web applications is concerned:

- The greater the connectivity the more time-consuming the development of the Web application becomes (confirms hypothesis H2).
- The higher the p-compactness the more time-consuming the development of the Web application becomes (confirms hypothesis H3).

Table 2. Spearman's correlation coefficient for metrics vs. effort

Tasks	Hyperdoc. size	Connectivity	P-Compactness	P-Stratum
Createt	0.60**	0.81**	0.62**	0.29
Createg	0.12	0.2.4	0.05	0.01
Createv	nc	nc	nc	nc
Createa	-1.0	-1.0	-1.0	-1.0
Createc	0.61**	0.72**	0.61**	0.10
Createl	0.35	0.6.4**	0..41*	0.27
Structap	0..40*	0.63**	0.06	0.17
Designip	0.31	0..45*	0.20	0.30
Designii	0.25	0..46*	0.16	0.16
Testl	0.70**	0.56**	0..43*	0.22
Testm	0.50	0.87	nc	0.87

** denotes that the result is statistically significant at the 1% level nc - could not be computed
* denotes that the result is statistically significant at the 5% level

Table 3. Spearman's correlation coefficient for metrics vs. effort

Tasks	Hyperdoc. size	Connectivity	P-Compactness	P-Stratum
Createt	0.39	0.13	0.05	-0..42*
Createg	-0.27	0..42	0.6.4**	-0.23
Createv	nc	nc	nc	nc
Createa	nc	nc	nc	nc
Createc	0.18	0.19	0.15	-0.39
Createl	0.18	0..45*	0.06	-0.35
Structap	0.10	0.12	0.36	-0.27
Designip	0.26	0.59**	0..40	-0.06
Designii	0.38	0.63**	0.82**	0.27
Testl	-0.07	0.2.4	0..46*	-0.20
Testm	-1.0	0.50	0.87	0.00

** denotes that the result is statistically significant at the 1% level nc - could not be computed
* denotes that the result is statistically significant at the 5% level

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The Mann-Whitney U Test was used to check whether the LEL and HEL groups were part of the same population. The results showed that they were, which allowed us to merge the answers into one, as follows:

Table 4. Results for the independent variables

Development effort of Web applications		
Hypothesis	Metric	Result
H1	Hyperdocument size	Confirmed
H2	Connectivity	Confirmed
H3	P-Compactness	Confirmed
H4	P-Stratum	Not enough evidence

A replication of the case study would help to confirm whether or not stratum could be used as a measure of deveOopment effort.

3. Conclusions

This paper has reported the findings of a quantitative case study aimed at empirically validating proposed metrics. The case study involved the development of Web applications using the principles of the Cognitive Flexibility Theory. The data was collected using questionnaires, which were answered by the subjects who participated in the case study.

Results have shown that three of the four metrics proposed – hyperdocument size, connectivity and p-compactness - have presented statistically significant correlations with development effort, suggesting that these metrics could be used in the future as parameters of an effort prediction model of Web development.

An important lesson learnt from the use of measurement is that it is better to start from a set of metrics addressing important improvement areas, and evolve these metrics over time, instead of debating forever, trying to find perfect metrics [6]. We believe that the steps used to propose and validate the metrics were as important as, or even more important than, the metrics themselves.

As part of our future work we are currently developing a tool that measures our proposed metrics automatically. Our aim is to use this tool in our next evaluation, which will be a replication of the case study presented in this paper. We hope that the results obtained from the replication study will give us the necessary feedback towards the definition of an effort prediction model of Web development.

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APPENDIX A - QUESTIONNAIRE

Questionnaire:

Name:
Surname:

For all time-related questions, please supply your estimate of elapsed time, in hours, considering only the mechanical aspects of the hypermedia design tasks and not the time it took you to research the contents. The underlined terms are defined in the appendix.

---

1) How many web pages does your application have? _____
2) How many web pages did you develop from scratch? _____
3) How many web pages did you reuse from other sites? _____

4) If you reused a web page, did you do this by (tick more than one if necessary):

   making a link to it from your application ( )
   making a local copy of that page and using it ( )

5) When reusing a web page did you (tick more than one if necessary):

   add new links to it ( )
   not add new links to it ( )

6) Approximately how many links did you create? _____

7) What is the <u>compactness</u> of the application?

Completely Connected	5	4	3	2	1	Completely Disconnected

8) What is the <u>stratum</u> of the application?

High stratum	5	4	3	2	1	No stratum

9) Please circle the structure which best describes how the core documents of the application are organised:

   <u>Sequence</u> <u>Hierarchy</u> <u>Network</u>

12) For each of the following tasks, please estimate the time it took to:

   Create all:

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a) text contents _____
b) graphics contents _____
c) video contents _____
d) audio contents _____

Create all:
e) links_____

Structure the application _____

Design the Interface:
h) planning of the interface _____
i) implementing of the interface_____

Test the application:
j) testing the links in the application_____
k) testing media, i.e., testing if the video works, the sound works, etc. _____

h) What tools did you use to develop the application and for what purpose? Tick more than one if appropriate:

	Purpose
( ) An HTML editor	_____
( ) An application generator	_____
( ) Software to organise and manage the HTML files	_____
( ) if other, which one?	_____

APPENDIX

Compactness - the connectedness of the application. A high compactness indicates that from any document a user can easily reach, using links, any other document in the application, suggesting a large amount of cross-referencing.

Hierarchy - a structure of an application where the documents are organised as a tree (see figure 1).

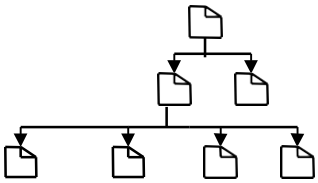

Figure 1 - A Hierarchy

Network - application where its structure is arranged as a net of documents and links (see figure 2).

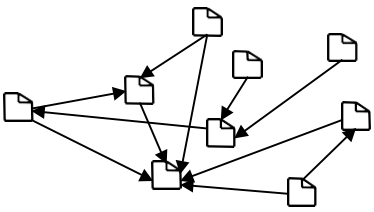

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Figure 2 - A Web

Sequence - a structure of an application where the documents are linearly linked (see figure 3).

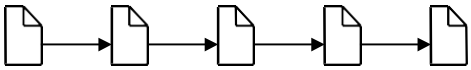

Figure 3 - A sequence

Stratum - suggests if there is an order for reading the application. Minimum stratum means that structurally it makes no difference from what document one starts reading. Maximum stratum is achieved in a linear application (the topology is a sequence).

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